Non-Conductive Piezoelectric Motor

Cost Analysis per Unit — CSMFAB 020

Category: Aegis Home — Motors **Unit Basis:** 1 standard 500 N·m motor unit (10cm diameter) **Date:** June 2026 | **Document Type:** CSMFAB Cost Study V2.0

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1. Product Summary

This cost analysis is derived from the V2.0 specification and v1to2 versioning document for this product. All material costs reflect 2026 market pricing. Three production scales are costed: - **Prototype:** 1-5 units (single-setup, high labor, no tooling amortization) - **Small Batch:** 100-500 units/year (semi-tooled, moderate labor) - **Production Volume:** 5,000-50,000 units/year (fully tooled, optimized labor)

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2. Bill of Materials (BOM)

Component	Quantity/Unit	Unit Price	Extended Cost
KNbO3-BaTiO3 ultrasonic stator elements (12 pc)	0.3 kg	$180.00/kg	$54.00
ZrO2 ceramic rotor disc	0.8 kg	$250.00/kg	$200.00
Si3N4 ceramic output shaft bearings (set)	0.3 kg	$90.00/kg	$27.00
PEEK CF40 housing	0.6 kg	$280.00/kg	$168.00
PMMA optical fiber control wiring	0.1 kg	$25.00/kg	$2.50
KNbO3-BaTiO3 driver electronics (GIC-immune DSP)	0.2 kg	$350.00/kg	$70.00
Aerogel thermal isolation pad (motor base)	0.1 m²	$250.00/kg	$25.00
BOM Subtotal			**$546.50**

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3. Labor, Process & Overhead Costs

Cost Element	Prototype	Small Batch	Production Volume
BOM Materials	$546.50	$448.13	$355.23
Direct Labor	$680.00	$460.00	$280.00
Overhead (15% of labor)	$102.00	$69.00	$42.00
Quality Control & Testing	$81.60	$36.80	$14.00
Packaging & Logistics	$25.00	$20.00	$15.00

Tooling & Equipment Notes: Piezo stator fabrication NRE per design: $35,000. High repeatability once tooled.

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4. Per-Unit Cost Summary

Scale	Total COGS	Suggested MSRP Range	Gross Margin
Prototype (1-5 units)	**$2,580**	$4,128 – $5,676	35-55%
Small Batch (100-500/yr)	**$1,820**	$2,730 – $3,640	33-50%
Production Volume (5K-50K/yr)	**$1,280**	$1,792 – $2,304	28-45%

Market Reference: $1,500-2,200 (ultrasonic motor market: this torque level ~$3,000-8,000; competitive at volume)

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5. Key Cost Drivers (V2.0)

Driver	Impact	Mitigation
ZrB2/Si3N4 UHTC ceramics	HIGH — specialty powder $350+/kg	Bulk procurement; domestic synthesis pilot (Japan NIMS CO2 route)
MXene Ti3C2Tx	VERY HIGH — $2,000/kg at 2026 scale	Tile pattern minimizes usage; price projected -60% by 2028 as production scales
Elium® thermoplastic BFRP	MEDIUM — recyclability adds value; $12/kg resin	Phoenix Protocol recycling recovers 100% monomer; net cost offset
Tooling NRE	HIGH for first production run	Paid by pre-orders or grant funding; amortized quickly
Flash Sintering process	MEDIUM — shares SPS equipment	Regional ceramic sintering facility sharing model

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6. Phoenix Protocol Circular Economy Value

At end-of-life, material recovery adds back-value: - **Recovered material value per unit:** $98 – $191 - **Recovery reduces net lifecycle cost by:** 15-30% - **Critical minerals (In, Y, Co, Ti) recovered** via ionic liquid extraction

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End of Non-Conductive Piezoelectric Motor Cost Analysis | CSMFAB 020 | June 2026